



Hypercalcemia

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Introduction

Calcium is the most abundant cation found in the human body and its homeostasis is tightly regulated. Hypercalcemia is a common disorder that is most often asymptomatic due to gradual and mild elevations in serum calcium that are detected on routine blood work. The presence of malignancy should be suspected if hypercalcemia develops rapidly, calcium levels are very high or systemic symptoms are present. Primary hyperparathyroidism (PHPT) is the other common cause of increased serum calcium levels. Approximately 90% of patients with hypercalcemia have PHPT or hypercalcemia of malignancy.^[1] Diagnostic and treatment considerations are reviewed in this chapter.

Goals of Therapy

- Correct extracellular fluid (ECF) volume contraction and dehydration
- Normalize serum calcium levels
 - enhance renal excretion of calcium
 - inhibit accelerated calcium resorption from bone
 - decrease intestinal calcium absorption
- Alleviate signs and symptoms of hypercalcemia
- Diagnose and treat underlying disorder

Investigations

- History and physical examination^{[1][2]} with special attention to:
 - onset and duration of symptoms: anorexia, constipation, nausea, vomiting, altered mental status, drowsiness, malaise, hyporeflexia, polydipsia, polyuria, bone pain and muscle weakness are symptoms usually associated with acute hypercalcemia; chronic hypercalcemia is usually asymptomatic
- Laboratory evaluation:^{[1][2][3][4][5]}
 - serum calcium and albumin; corrected Ca^{++} value (mmol/L) = $(0.02 \times [40 - \text{measured albumin g/L}]) + \text{measured } \text{Ca}^{++}$ or serum ionized calcium where available
 - serum intact parathyroid hormone (PTH); perform this test before giving bisphosphonates, as they can alter serum PTH levels
 - serum parathyroid hormone-related peptide (PTHrP) and 1,25-dihydroxyvitamin D level if PTH low or low-normal
 - 24-hour urine collection to measure calcium-to-creatinine ratio (if familial hypocalciuric hypercalcemia or milk-alkali syndrome is suspected)
 - serum phosphate, alkaline phosphatase, total protein, serum creatinine and urea

Therapeutic Choices

- Aggressiveness of initial interventions depends on the rapidity of onset and severity of the hypercalcemia (see [Table 1](#)).
- Definitive therapy for long-term control of hypercalcemia requires diagnosis and treatment of the underlying condition (see [Figure 1](#)).

Table 1: Using the Duration of Hypercalcemia, Parathyroid Hormone and Parathyroid Hormone-Related Peptide Level to Suggest a Diagnosis

Duration of Hypercalcemia	PTH	PTHrP	Most Likely Diagnosis
Acute or Chronic	High or Normal	N/A ^[a]	• Primary hyperparathyroidism • Familial hypocalciuric hypercalcemia possible if urine calcium/creatinine ratio <0.02
Acute	Low	High or Normal	• Malignancy (perform a more complete clinical evaluation)

Duration of Hypercalcemia	PTH	PTHrP	Most Likely Diagnosis
	Low	Low	• Malignancy-related osteolytic bone lesion (perform a more complete clinical evaluation)
Chronic	Low	Low	• Adrenal insufficiency • Drug-induced, e.g., androgens, antiestrogens, derivatives and analogues of vitamin A, estrogens, excess vitamin A or D, ingestion of >3 g elemental calcium per day, lithium, parenteral nutrition, progestins, thiazide diuretics, sodium-glucose cotransporter 2 inhibitors, immune checkpoint inhibitors and rebound after denosumab discontinuation • Familial disorders • Granulomatous diseases • Hyperthyroidism or milk-alkali syndrome

^[a] If PTH is high, measurement of PTHrP is not required.

Abbreviations: N/A = not applicable; PTH = parathyroid hormone; PTHrP = parathyroid hormone-related peptide

Nonpharmacologic Choices

Mobilize

Hypercalcemia is exacerbated by immobilization. Ambulation in appropriate patients helps to reduce bone resorption and normalize serum calcium.^[1]

Diet

Dietary changes rarely correct hypercalcemia. Patients with sarcoidosis and other granulomatous causes of hypercalcemia or those who have vitamin D-mediated hypercalcemia may benefit from dietary calcium restriction. Limit the use of vitamin D supplements, calcium supplements or calcium-containing antacids in all patients.^{[1][4]}

Hemodialysis

If it is available and can be instituted promptly, hemodialysis (low or zero calcium bath) provides rapid correction of severe symptomatic hypercalcemia. It is of value in those with severe hypercalcemia and severe renal impairment or heart failure who could not safely tolerate hydration.^{[1][4]}

Pharmacologic Choices

General Measures

Stop Offending Agents

If possible, discontinue offending agents (see drug-induced causes in [Table 1](#)) and replace with agents that do not exacerbate hypercalcemia.

Replace Fluids

Patients with hypercalcemia are often dehydrated as a result of a renal water-concentrating defect induced by hypercalcemia and by decreased fluid intake resulting from nausea and vomiting. The resulting reduced glomerular filtration rate limits the ability of the kidney to excrete calcium. Expansion of intravascular volume enhances renal calcium clearance. Hydration alone usually reduces serum calcium by 0.25 to 0.375 mmol/L over the first 24 hours.^[6] This effect is transient unless additional treatments are instituted.^{[1][3][4][6]} Serum magnesium and potassium may also decrease with hydration and should be monitored and replaced as needed.

Patients with mild to moderate asymptomatic hypercalcemia should drink 3 L per day of noncaffeinated beverages to achieve and maintain euvoemia. Patients with moderate to severe symptomatic hypercalcemia should receive intravenous **normal saline** (0.9% NaCl) bolus of 1 to 2 litres and then a continuous infusion at 250–500 mL/hour, if safe to do so, until euvoemic and a urine output of 100–150 mL/h is maintained.^{[1][4][6][7]} Renal dysfunction is common among patients with hypercalcemia. Careful monitoring of hydration is essential, especially in older patients and those with pre-existing renal insufficiency or left ventricular systolic dysfunction, as fluid overload can develop. Refer to [Figure 1](#) for further information on the management of mild, moderate and severe hypercalcemia.

Use a loop diuretic (**furosemide**, **ethacrynic acid**) to treat fluid overload and heart failure as needed, but only after volume depletion is fully corrected. Loop diuretics can aggravate hypercalcemia by depleting extracellular fluid volume so routine use is discouraged. Thiazide diuretics are contraindicated because they impair calcium excretion.^[8]

Normalize Serum Phosphate

Hypophosphatemia, seen mostly in primary hyperparathyroidism and malignancy-associated hypercalcemia, exacerbates hypercalcemia by increasing renal synthesis of 1,25-dihydroxyvitamin D, which consequently reduces bone formation, increases bone resorption and increases intestinal

absorption of dietary calcium. **Oral phosphates** (1–2 g elemental phosphate/day) can be given safely in all patients, except those with renal insufficiency, to increase serum phosphate to the low normal range (0.8–1 mmol/L). Though rare, calcium phosphate can precipitate and cause serious organ damage (heart, kidney, lungs, blood vessels) if phosphate levels are increased to >1 mmol/L or if IV phosphate is administered.^{[9][10]}

Specific Measures

[Table 2](#) lists the drugs used in the management of hypercalcemia.

Hypercalcemia due to Primary Hyperparathyroidism

Surgery to remove the parathyroid glands remains the first-line measure for control of primary hyperparathyroidism in both symptomatic and asymptomatic patients.^{[11][12][13]} Medical management can be used for asymptomatic patients who do not meet established criteria for surgery.^[13] Adequate hydration should be recommended and calcium intake should not be restricted. Vitamin D deficiency should be corrected.^[14] Antiresorptive therapy with a **bisphosphonate** or **denosumab** is usually reserved for patients with moderate to severe hypercalcemia or those who exhibit osteopenia.^{[14][15]}

Cinacalcet is a calcimimetic that is recommended for the reduction of hypercalcemia in patients with parathyroid carcinoma and for the reduction of clinically significant hypercalcemia in patients with PH for whom parathyroidectomy is not clinically appropriate or is contraindicated.^{[16][17][18]} Cinacalcet is also indicated for treatment of secondary hyperparathyroidism in patients with chronic kidney disease who are on dialysis.

Hypercalcemia due to Malignancy

Anticancer therapy aimed at the underlying malignancy is the key to long-term calcium control, though not all patients are candidates. The bisphosphonates **clodronate**, **pamidronate** and **zoledronic acid** are effective for hypercalcemia of malignancy.^[19] They should be used along with cancer therapy, if appropriate, in patients with moderate symptomatic hypercalcemia (>3 mmol/L) and patients with severe hypercalcemia (>3.5 mmol/L with or without symptoms).^{[7][20][21]}

For acute management, parenteral zoledronic acid 4 mg as a 15-minute infusion is the preferred treatment due to ease of use and superior efficacy as compared to the 2–4 hr infusion of pamidronate.^{[22][23]} Reduction in calcium and improvement in symptoms with an IV bisphosphonate is generally seen within 24–72 hours with a complete response expected in 7–10 days. Bisphosphonates are usually given in conjunction with saline and if hypercalcemia is severe, they can be administered with calcitonin. For patients who respond, serum calcium levels can be maintained at acceptable levels for up to 5 weeks before re-treatment is required, with zoledronic acid exhibiting a longer duration of action versus pamidronate.^[6] Redosing is indicated if the response is inadequate after 7 days. Patients with multiple myeloma or metastatic breast, lung, or prostate cancer benefit from reduced bone pain, decreased skeletal morbidity, and prevention of hypercalcemia recurrence with prolonged therapy.^{[24][25]} Prolonging infusion time of zoledronic acid to 30–60 minutes and pamidronate to 4 hours may reduce the risk for renal toxicity

in patients with pre-existing renal dysfunction (ClCr <60 mL/min). Clodronate has a short duration of action and is more costly. There is insufficient evidence to recommend the use of other bisphosphonates (e.g., alendronate, risedronate) for serum calcium control.^[22] Premedication with **acetaminophen** 650 mg can help prevent bisphosphonate-induced fever.

Calcitonin rapidly reduces serum calcium levels in severe hypercalcemia; however, the response tends to abate within 48 hours and therefore is considered a bridging therapy.^[23] Calcitonin treatment should be limited to 48–72 hours due to tachyphylaxis. **Gallium** is not available for use in Canada.

Denosumab, a monoclonal antibody that binds to the bone resorption mediator RANKL, can be used as an alternative to bisphosphonate therapy and for hypercalcemia of malignancy that is refractory to IV bisphosphonate therapy. Denosumab can cause severe hypocalcemia in patients with an estimated glomerular filtration rate (eGFR) less than 30 mL/min/1.73 m², including patients receiving kidney replacement therapy.

Adjunctive corticosteroid therapy, e.g., **prednisone** 40–100 mg/day or equivalent for up to 1 week, may be useful in patients with lymphoma, myeloma, lymphoid leukemia and breast cancer. Corticosteroids inhibit osteoclastic bone resorption by decreasing tumor production of locally active cytokines, in addition to having direct tumorolytic effects. Additionally, corticosteroids decrease calcitriol production and thus intestinal calcium absorption.

Hypercalcemia due to Granulomatous Diseases

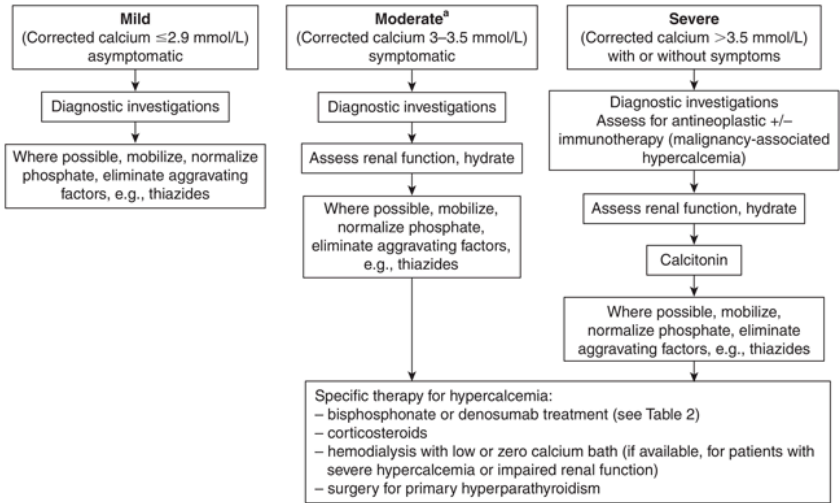
Excess production of 1,25-dihydroxyvitamin D is characteristic of granulomatous diseases, e.g., sarcoidosis. Therapy includes restricting vitamin D and calcium intake and avoiding excessive exposure to sunlight. Corticosteroids may be used in acutely ill patients.^[1]

Therapeutic Tips

- Overly aggressive use of loop diuretics can aggravate hypercalcemia by depleting extracellular fluid volume. Routine use of loop diuretics has not been shown to be beneficial and is discouraged.^[8]
 - To reduce serum calcium rapidly (within 6–12 hr) in severe hypercalcemia of malignancy, use **calcitonin** plus hydration. Serum calcium usually declines 0.8 mmol/L at 12–24 hours following combined therapy. To augment and prolong serum calcium control, use definitive therapy, e.g., antineoplastic therapy, for the underlying cause. Until cancer therapy takes effect, or if it is not an option, use a **bisphosphonate** when dehydration is corrected and adequate urine output is achieved—preferably within 24 hours of a hypercalcemia diagnosis.
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Algorithms

Figure 1: Management of Hypercalcemia



^[a] Asymptomatic patients with hypercalcemia classified as “moderate” should be managed in the same way as mild, asymptomatic patients.

Drug Table

Table 2: Drugs Used for the Treatment of Hypercalcemia

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Antiparathyroid Hormones			
calcitonin salmon Calcimar, generics	Initial: 4 units/kg Q12H SC or IM If unsatisfactory response after 1–2 days, may increase to 8 units/kg Q12H; may increase to a maximum of 8 units/kg Q6H	Nausea, vomiting (dose dependent), flushing of face and hands. Hypersensitivity reaction.	Used for rapid early effect, i.e., within 4–6 h. Duration: 1–4 days. Tachyphylaxis develops in 2–7 days; combining with corticosteroids may increase efficacy and reduce tachyphylaxis.
\$60/400 units vial			

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Bisphosphonates			
clodronate  , oral Clasteon	1600–2400 mg PO given once daily or divided BID (2 h AC or 2 h PC)	Nausea, vomiting, anorexia, diarrhea, muscle cramps. Osteonecrosis of the jaw (see Comments).	Use with bisphosphonate lowers serum Ca ⁺⁺ levels more rapidly than with either agent alone. For patients with suspected sensitivity perform skin test; if marked erythema and/or wheal formation does not occur within 15 min after administration, therapy can begin.
\$5/4 capsules of 400 mg	Maximum daily dose is 3200 mg; ensure adequate fluid rehydration		Concomitant intake of oral iron, calcium, magnesium, aluminum, e.g., antacids, may prevent absorption of oral clodronate. Do not take with or within 1 h before or after food or milk. Pamidronate or zoledronic acid are preferred agents. Osteonecrosis of the jaw: regular dental exam, especially prior to therapy or within 3 months, if possible, as well as good oral hygiene.
pamidronate  , generics	Tumor-induced hypercalcemia: adequate hydration with normal saline necessary before treatment	Local infusion site reactions, febrile reaction within 24–48 h of infusion; may be accompanied by	Onset: 24–72 h; maximum lowering occurs in 3–7 days. Duration: 3–4 wk.
\$275/90 mg vial			

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
	30–90 mg IV, depending on serum Ca ⁺⁺ ; consult product monograph for details	malaise, rigor, fatigue and flushes. Posterior uveitis (rare).	Repeat infusion if Ca ⁺⁺ level does not fall within 7 days or if hypercalcemia recurs.
	Not to exceed 90 mg in 250 mL administered over 2–4 h	Osteonecrosis of the jaw (see Comments).	Monitor renal function (serum creatinine and/or urea), evaluate within 1 wk after infusion and just prior to next scheduled dose.
	Total dose can be given as a single infusion or divided into smaller infusions given over a period of 2–4 consecutive days		Case reports of safe and effective use, without dosage modification, in patients on hemodialysis.
	Do not dilute with calcium-containing solutions or give as bolus injections		Infrequent osteonecrosis of the jaw: regular dental exam, especially prior to therapy or within 3 months, if possible, as well as good oral hygiene.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
zoledronic acid Aclasta, Zometa Concentrate, generics \$420/4 mg vial	Tumor-induced hypercalcemia: establish and maintain adequate hydration 4 mg in patients with albumin-corrected serum Ca ⁺⁺ ≥3 mmol/L given as a single dose IV infusion over 15 min ^[18] Re-treatment: 8 mg as a single dose over 15 min, in patients demonstrating complete or partial response without normalization of Ca ⁺⁺ levels; use only in patients who can tolerate standard rehydration procedures (3–5 L fluids and >400 mmol of NaCl per day) Do not dilute with calcium-containing solutions or give as bolus injections	Febrile reactions, nausea, vomiting, flulike syndrome, local infusion site reactions, mild hypomagnesemia, hypokalemia and hypophosphatemia. Osteonecrosis of the jaw (see Comments). Renal function deterioration reported with infusion <15 min.	Onset: 24–72 h. Duration: 4–5 wk. Repeat infusion if Ca ⁺⁺ level does not fall within 7 days or if hypercalcemia recurs. Monitor renal function, evaluate within 1 wk after infusion and just prior to next scheduled dose. Infrequent osteonecrosis of the jaw: regular dental exam, especially prior to therapy or within 3 months, if possible, as well as good oral hygiene.

Drug Class: Calcimimetics

cinacalcet Sensipar, generics \$20/60 mg tablet	Dialysis patients: Initial: 30 mg once daily PO; titrate every 2–4 wk to a maximum dose of 180 mg once daily to achieve a target PTH level Primary hyperparathyroidism: Initial: 30 mg BID PO; titrate every 2–4 wk to 60 mg, then 90 mg BID to reduce serum Ca ⁺⁺ levels	Nausea, vomiting, diarrhea, myalgia, hypocalcemia.	Inhibits CYP2D6; dose adjustments may be required for concomitant CYP2D6-metabolized drugs with narrow therapeutic index, e.g., tricyclic antidepressants, flecainide. Increases the sensitivity of the Ca ⁺⁺ -sensing receptor to circulating serum Ca ⁺⁺ , thereby reducing the secretion of PTH .
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Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
	Moderate to severe hepatic impairment increases drug concentrations by approximately 2- to 4-fold; dosage modification may be required		Onset: PTH level reduced within 2–6 h after a dose. Food increases bioavailability. Monitor serum Ca ⁺⁺ , serum PO ₄ and PTH .
Drug Class: Corticosteroids, systemic			
dexamethasone generics \$24/40 tabs of 4 mg	8 mg BID PO for 5–10 days		Slow onset. Potential for hyperglycemia, osteoporosis, increased susceptibility to infection.
hydrocortisone sodium succinate Solu-Cortef \$20/500 mg vial	200–400 mg daily IV for 3–5 days		Slow onset. Potential for hyperglycemia, osteoporosis, increased susceptibility to infection.
prednisone generics \$2/100 tabs of 5 mg	40–60 mg daily PO for 5–10 days		Slow onset. Potential for hyperglycemia, osteoporosis, increased susceptibility to infection.
Drug Class: Crystalloids			
sodium chloride 0.9% (normal saline, NS) generics \$7/1000 mL bag	Initial bolus of 1–2 L, then 250–500 mL/h IV to maintain urine output at 100–150 mL/h Dependent on concomitant disease states and extent of hypercalcemia		Onset: immediate. Can be given ± other electrolytes. Contraindications: renal insufficiency, heart failure. Carefully assess for volume overload.

Drug/Cost ^[a]	Dosage	Adverse Effects	Comments
Drug Class: Diuretics, loop			
furosemide  generics \$4/40 mg vial	Initial: 40–80 mg IV Q1–4H after volume expansion	Electrolyte abnormalities, e.g., hypokalemia, orthostatic hypotension.	Given after aggressive rehydration has been used to restore intravascular volume. Prevents fluid overload and inhibits calcium reabsorption in the distal renal tubule.
Drug Class: RANK Ligand Inhibitors			
denosumab Xgeva, Wyost \$340/120 mg vial	120 mg every 4 wk SC	Hypersensitivity reactions including anaphylaxis. Atypical femoral fractures. Osteonecrosis of the jaw (see Comments).	Onset: 3–10 days. Time to complete response: 23 days. Median duration response: 104 days. Good oral hygiene and regular dental exam recommended to avoid or detect osteonecrosis of the jaw. Do not use in combination with bisphosphonates. Can cause severe hypocalcemia in patients with an eGFR <30 mL/min/1.73 m ² , including patients receiving kidney replacement therapy.
^[a] Drug cost only.  Dosage adjustment may be required in renal impairment; see Dosage Adjustment in Renal Impairment . Abbreviations: eGFR = estimated glomerular filtration rate; PTH = parathyroid hormone; RANK = receptor activator of nuclear factor kappa-B			

Suggested Readings

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